

FLORIANOPOLIS, Brazil

WMO: 838990

Lat: 27.670S

Lon: 48.547W

Elev: 6

StdP: 101.25

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	10.0	3.2	4.8	12.4	5.1	5.5	12.8	8.7	17.8	7.9	17.4	1.8	300	0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.9	32.2	25.3	31.0	25.0	30.0	24.5	26.5	30.1	25.9	29.3	25.4	28.5	4.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.5	27.9	25.0	20.1	27.6	24.2	19.2	26.9	82.8	30.4	80.2	29.7	77.8	28.4	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.4	7.5	6.8	DB	5.0	35.7	2.1	1.8	3.5	36.9	2.3	38.0	1.1	39.0	-0.5	40.3
			WB	3.9	27.7	1.8	0.7	2.7	28.3	1.6	28.7	0.6	29.1	-0.6	29.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.4	25.5	25.6	24.6	22.6	19.8	17.9	17.0	18.0	18.9	20.8	22.3	24.4	(6)
	DBStd	3.92	2.07	2.10	2.12	2.45	2.58	2.99	3.17	3.08	2.63	2.28	2.26	2.35	(7)
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)
	HDD18.3	195	0	0	0	2	15	43	62	45	23	4	1	0	(9)
	CDD10.0	4173	479	436	453	379	302	238	218	249	268	335	368	446	(10)
	CDD18.3	1327	221	203	195	131	59	31	22	36	42	81	119	187	(11)
	CDH23.3	7248	1621	1500	1238	599	151	76	58	94	87	204	475	1144	(12)
(13)	CDH26.7	1699	435	406	297	105	18	9	10	21	10	24	78	287	(13)
(14)	Wind	WSAvg	3.2	3.2	3.0	2.9	2.8	2.9	3.0	3.5	3.7	3.6	3.7	3.6	(14)
(15) Precipitation	PrecAvg	1602	222	205	171	119	101	76	88	78	144	148	136	146	(15)
	PrecMax	2361	625	536	325	314	310	245	186	262	307	339	595	340	(16)
	PrecMin	827	29	53	50	34	3	7	4	10	18	36	36	9	(17)
	PrecStd	366	109	107	70	64	67	46	46	55	72	69	93	73	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.0	32.9	30.9	28.9	28.0	28.1	29.4	27.9	29.0	31.0	33.7	(19)
		MCWB	26.2	26.2	25.8	24.7	22.6	20.9	20.1	20.5	21.6	23.0	24.1	25.7	(20)
	2%	DB	31.9	32.0	31.0	29.1	26.2	25.2	24.8	25.7	25.7	27.0	28.8	31.1	(21)
		MCWB	25.4	25.5	25.3	23.7	21.3	20.4	19.6	20.1	20.6	21.8	22.9	24.8	(22)
	5%	DB	30.2	30.6	29.8	27.9	25.0	23.8	22.9	23.9	24.0	25.5	27.1	29.6	(23)
		MCWB	24.9	25.2	24.7	22.9	20.8	19.6	18.9	19.3	19.9	21.1	22.0	24.1	(24)
	10%	DB	29.0	29.1	28.2	26.3	23.8	22.1	21.1	22.1	22.5	24.1	26.0	28.1	(25)
		MCWB	24.4	24.5	23.7	22.4	20.2	18.9	18.0	18.5	19.0	20.4	21.4	23.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.0	26.9	25.7	23.7	22.3	21.2	21.6	22.2	23.9	25.0	26.7	(27)
		MCDB	30.9	31.1	30.5	29.0	27.0	25.8	25.5	26.4	26.2	27.6	29.2	30.9	(28)
	2%	WB	26.4	26.3	26.1	24.7	22.5	21.1	20.0	20.6	21.2	22.7	23.8	25.7	(29)
		MCDB	29.9	29.9	29.3	27.6	25.1	23.9	23.4	24.5	24.0	25.3	27.0	29.2	(30)
	5%	WB	25.7	25.7	25.4	24.0	21.6	20.3	19.3	19.7	20.6	22.0	23.1	24.9	(31)
		MCDB	29.0	29.1	28.3	26.5	23.9	22.8	22.0	22.8	22.9	24.4	25.7	28.1	(32)
	10%	WB	25.2	25.2	24.7	23.4	20.8	19.5	18.6	19.2	19.8	21.2	22.5	24.2	(33)
		MCDB	28.0	28.2	27.3	25.6	22.9	21.8	20.8	21.7	21.8	23.3	24.9	27.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	6.9	7.0	7.2	7.3	7.4	7.4	7.5	6.6	6.2	6.8	6.9	(35)
		MCDBR	8.5	8.5	8.2	8.3	8.4	8.7	9.1	9.0	8.1	7.8	8.3	8.8	(36)
	5% WB	MCWBR	3.7	3.7	3.7	3.9	4.2	4.3	4.6	4.3	4.1	4.0	4.0	4.0	(37)
		MCWBR	7.8	7.8	7.6	7.1	7.5	8.0	8.2	8.2	7.2	6.9	7.4	7.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.418	0.413	0.400	0.397	0.370	0.391	0.399	0.469	0.535	0.466	0.424	0.420	(40)
		taud	2.438	2.452	2.481	2.464	2.512	2.441	2.389	2.138	1.952	2.217	2.366	2.405	(41)
	Ebn at Noon		926	914	895	848	826	778	784	761	757	855	915	926	(42)
		Edh at Noon	123	118	109	101	87	90	98	137	181	147	131	127	(43)
(44) All-Sky Solar Radiation	RadAvg		5.88	5.52	4.80	3.91	3.02	2.62	2.81	3.39	3.85	4.47	5.50	5.86	(44)
	RadStd		0.33	0.44	0.40	0.36	0.32	0.18	0.31	0.35	0.43	0.58	0.52	0.48	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page